

Prescribed-Time Cooperative Tracking for Second-Order UAVs: A Fully Distributed Backstepping Approach

Feiyang Chen, Xingxing Ju

Abstract—Cooperative dynamic target tracking for multi-UAV systems requires rapid and decentralized coordination. Existing distributed optimization strategies often suffer from conservative convergence bounds and rely on global network topology, hindering practical deployment. To overcome these limitations, we propose a fully distributed prescribed-time optimization framework specifically designed for second-order UAV dynamics. First, a novel adaptive gain mechanism utilizing solely local errors eliminates the dependence on global Laplacian eigenvalues. Second, a backstepping controller with analytic cross-cancellation terms is developed to decouple position-velocity dynamics, successfully overcoming the infinite-gain numerical singularity inherent in prescribed-time control. Finally, a state-switching mechanism enables strict piecewise convergence: achieving exact spatial formation assembly within a user-specified time T , followed by a seamless transition to asymptotic optimal tracking without steady-state errors. Simulations validate the framework's collision-free safety and control energy economy, providing a highly practical decentralized solution for dynamic UAV swarm deployments.

Index Terms—UAV swarm, target tracking, distributed optimization, prescribed time, backstepping control.

I. INTRODUCTION

IN recent years, cooperative control of multi-UAV systems has demonstrated potential in time-critical missions such as disaster rescue, power line inspection, and battlefield surveillance [1]–[4]. Among these tasks, cooperative dynamic target tracking requires the UAV swarm to continuously optimize its geometric encirclement configuration in real time to intercept or monitor maneuvering targets [5]–[10]. Consequently, distributed time-varying optimization has emerged as an effective mathematical framework for solving such cooperative tracking problems [11]–[16].

In practical tactical operations, convergence speed is a critical metric. Traditional distributed optimization algorithms

rely on asymptotic convergence, theoretically requiring infinite time to reach the optimal state. While finite- and fixed-time control techniques have been introduced to address this [17]–[19], their convergence upper bounds are often overly conservative and heavily rely on complex control parameters. In time-sensitive real-world scenarios, such as emergency search-and-rescue or tactical encirclement, mission specifications dictate that the UAV swarm must *complete tactical deployment and formation assembly within a predefined deadline*. Therefore, shifting from fixed-time to prescribed-time control is driven by engineering requirements rather than theoretical curiosity [20]–[22].

By introducing a time-scaling function with a temporal singularity, prescribed-time control forces the system states to converge within a user-specified time T . While prescribed-time algorithms have recently advanced distributed optimization for multi-agent systems [23]–[32], their direct application to complex UAV swarms faces two critical engineering bottlenecks.

First, most existing protocols require prior knowledge of global topological information, such as the minimum non-zero Laplacian eigenvalue λ_2 [17]. Relying on such global parameters is often physically unfeasible for UAV swarms with limited communication bandwidth. Although recent works proposed fully distributed prescribed-time algorithms without topological dependence [33], they are restricted to idealized first-order integrator models.

Second, unlike first-order models, actual UAVs are governed by second-order dynamics characterized by deep position-velocity coupling [34], [35]. Under prescribed-time control, the “infinite gain” phenomenon near the specified time T can amplify transient fluctuations through the double-integrator chain, leading to numerical singularities, actuator saturation, and system instability. Decoupling these dynamics under a fully distributed architecture remains a critical technical gap.

Motivated by the aforementioned practical challenges, this article proposes a fully distributed prescribed-time optimization control strategy for cooperative UAV target tracking. The main contributions are summarized as follows:

- **Fully Distributed Framework for Second-Order UAV Dynamics:** By designing edge-based and node-based adaptive gains that rely exclusively on local interactions, the proposed framework eliminates the restrictive dependence on global network topologies, such as Laplacian eigenvalues, enabling decentralized deployment for practical second-order UAV systems.

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- **Decoupling and Singularity Avoidance via Backstepping:** To address the deep position-velocity coupling inherent in actual UAV dynamics, a backstepping controller with analytic cross-cancellation terms is developed. This structural design overcomes the “infinite-gain” numerical singularity—a bottleneck in existing time-scaling prescribed-time controls—thereby guaranteeing bounded and realizable control inputs for UAV actuators.
- **Piecewise Convergence with Collision-Free Hard Constraints:** A state-switching mechanism integrates Artificial Potential Fields with the tracking dynamics. This creates a piecewise convergence strategy: the UAV swarm is forced to complete a collision-free spatial formation assembly within the user-specified time T , and subsequently transitions to asymptotic tracking of the time-varying global optimal trajectory without steady-state errors.

The main mathematical notations used throughout this article are summarized in Table I.

TABLE I. Notations

Notation	Description
$\mathbb{R}^m, \mathbb{R}^{N \times N}$	m -dimensional Euclidean space and $N \times N$ real matrices
$\mathbb{R}_{\geq 0}, \mathbb{R}_{> 0}$	Sets of non-negative and positive real numbers
$\mathbf{I}_N, \mathbf{1}_N, \mathbf{0}_N$	$N \times N$ identity matrix, N -dimensional column vector with all elements being 1 and 0
$\otimes, \ \cdot\ $	Kronecker product and Euclidean 2-norm
$\lambda_{\min}(\cdot), \lambda_{\max}(\cdot)$	Minimum and maximum eigenvalues of a real symmetric matrix
$\nabla f_i, \nabla^2 f_i$	Gradient vector and Hessian matrix of the local cost function f_i
$\mathcal{G}, \mathcal{V}, \mathcal{E}$	Undirected communication graph, set of nodes, and set of edges
$\mathcal{A}, L$	Adjacency matrix and Laplacian matrix of graph $\mathcal{G}$
$\mathcal{N}_i$	The neighbor set of node i
p_i, v_i, u_i	Physical position, velocity, and actual control input of the i -th UAV
$x_i, \delta_i(t)$	Virtual cooperative state and formation offset vector of the i -th UAV
$T, \rho(t)$	User-specified prescribed time and time-scaling function
$\alpha_{ij}(t), \beta_i(t)$	Edge-based and node-based distributed adaptive gains
$\Phi_i, d_i(t)$	Artificial potential field gradient and unknown environmental disturbance

II. PROBLEM FORMULATION AND PRELIMINARIES

This section models the UAV dynamics and formulates the cooperative tracking problem. The inherent challenges of prescribed-time missions are then analyzed to motivate the controller design, concluding with necessary mathematical preliminaries.

A. UAV Dynamics and Physical Characteristics

Unlike idealized first-order models, actual UAVs are governed by second-order cascaded dynamics which are intuitively

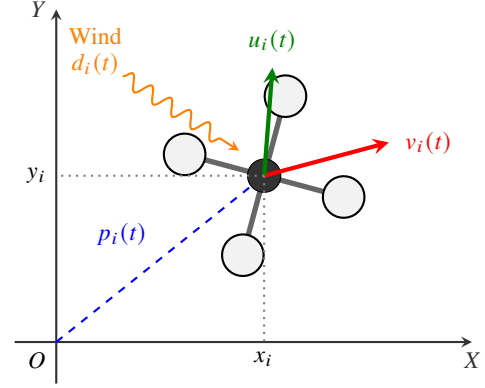


Fig. 1: The 2D dynamic model of the i -th UAV.

illustrated in Fig. 1. After feedback linearization, the motion of the i -th UAV ($i \in \{1, 2, \dots, N\}$) is described as:

$$\begin{cases} \dot{p}_i(t) = v_i(t) \\ \dot{v}_i(t) = u_i(t) + d_i(t) \end{cases}$$

where $p_i, v_i \in \mathbb{R}^m$ denote the spatial position and velocity, respectively. The term $d_i(t) \in \mathbb{R}^m$ represents the bounded lumped environmental disturbance satisfying $\|d_i(t)\| \leq D_{\max}$.

Due to the physical limits of UAV rotors, the actual control input $u_i(t) \in \mathbb{R}^m$ is subject to a strict actuator saturation constraint:

$$u_i(t) = \text{sat}(u_{ic}(t)) = \begin{cases} u_{ic}(t), & \|u_{ic}(t)\| \leq U_{\max} \\ U_{\max} \frac{u_{ic}(t)}{\|u_{ic}(t)\|}, & \|u_{ic}(t)\| > U_{\max} \end{cases} \quad (1)$$

where $u_{ic}(t)$ is the commanded ideal acceleration and $U_{\max} > 0$ is the maximum thrust limit.

This bounded second-order model highlights two critical engineering challenges: deep position-velocity coupling and vulnerability to actuator saturation. Specifically, the high-gain nature of prescribed-time control tends to generate excessive transient commands that easily violate the hard constraint (1), destabilizing the system. These physical limitations strictly dictate the decoupled *backstepping* control architecture developed in Section III.

B. Problem Formulation of Cooperative Target Tracking

To achieve dynamic encirclement and formation reconfiguration around the maneuvering target, a smoothly time-varying offset vector $\delta_i(t) \in \mathbb{R}^m$ is introduced. The virtual cooperative state $x_i(t)$ is defined as:

$$x_i(t) = p_i(t) - \delta_i(t)$$

Achieving consensus in the virtual state space, i.e., $x_i(t) = x_j(t)$ for all $i, j \in \mathcal{V}$, naturally guarantees the prescribed physical formation $p_i(t) - p_j(t) = \delta_i(t) - \delta_j(t)$.

Let $f_i(x_i, t)$ be the local time-varying cost function assigned to the i -th UAV. The cooperative target tracking task is cast as a distributed optimization problem subject to an equality manifold constraint:

Problem 1:

$$\begin{aligned} \min_{x_1, \dots, x_N \in \mathbb{R}^m} \quad & \sum_{i=1}^N f_i(x_i, t) \\ \text{s.t.} \quad & x_1 = x_2 = \dots = x_N \end{aligned} \quad (2)$$

Let $x^*(t) \in \mathbb{R}^m$ denote the exact time-varying optimal trajectory that globally minimizes problem (2). To formalize the tracking performance, the spatial consensus error $e_{cij}(t)$ and the optimal tracking error $e_{ti}(t)$ are defined as:

$$\begin{cases} e_{cij}(t) = x_i(t) - x_j(t) \\ e_{ti}(t) = x_i(t) - x^*(t) \end{cases}$$

To ensure inter-UAV collision avoidance during dynamic tracking, a state-based Artificial Potential Field (APF) is utilized. Let the relative position vector between nodes i and j be defined as $p_{ij} = p_i - p_j$. The repulsive potential function $U_{ij}(\|p_{ij}\|)$ is designed as:

$$U_{ij}(\|p_{ij}\|) = \begin{cases} \frac{1}{2} \left(\frac{1}{\|p_{ij}\| - d_{\min}} - \frac{1}{R - d_{\min}} \right)^2, & d_{\min} < \|p_{ij}\| \leq R \\ 0, & \|p_{ij}\| > R \end{cases}$$

where $d_{\min} > 0$ is the absolute safe distance acting as a physical hard constraint, and $R > d_{\min}$ is the detection radius. Note that $U_{ij} \rightarrow \infty$ as $\|p_{ij}\| \rightarrow d_{\min}^+$. The total repulsive gradient force exerted on UAV i by its neighbors is denoted as $\Phi_i = \sum_{j \in \mathcal{N}_i} \nabla_{p_i} U_{ij}(\|p_{ij}\|)$.

While subsequent theoretical derivations assume an obstacle-free airspace to maintain focus on the core prescribed-time tracking mechanisms, the proposed framework exhibits inherent environmental extensibility. By simply modeling external obstacles as virtual repulsive nodes k , their collision avoidance gradients $\nabla_{p_i} U_{ik}(\|p_i - p_k\|)$ can be directly superimposed onto the virtual velocity law without compromising the closed-loop Lyapunov stability.

Consequently, the ultimate control objective is to design a fully distributed protocol $u_i(t)$ such that $\lim_{t \rightarrow (t_0+T)^-} \|e_{cij}(t)\| = 0$ and $\lim_{t \rightarrow \infty} \|e_{ti}(t)\| = 0$, while strictly satisfying the collision avoidance constraint $\|p_{ij}(t)\| > d_{\min}$ and the actuator saturation limit $\|u_i(t)\| \leq U_{\max}$ throughout the dynamic process.

C. Graph Theory

The communication topology of the UAV swarm can be modeled by an undirected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{A})$, where $\mathcal{V} = \{1, 2, \dots, N\}$ is the set of nodes and $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ is the set of edges. $\mathcal{A} = [a_{ij}] \in \mathbb{R}^{N \times N}$ is the adjacency matrix, where if UAV i and j can exchange information, then $(j, i) \in \mathcal{E}$ and $a_{ij} = a_{ji} > 0$; otherwise, $a_{ij} = 0$. The neighbor set of node i is denoted as $\mathcal{N}_i = \{j \in \mathcal{V} \mid (j, i) \in \mathcal{E}\}$. The Laplacian matrix of graph $\mathcal{G}$ is defined as $L = \mathcal{D} - \mathcal{A}$, where $\mathcal{D} = \text{diag}\{d_1, \dots, d_N\}$ is the degree matrix with $d_i = \sum_{j \in \mathcal{N}_i} a_{ij}$.

Assumption 1: The communication topology graph $\mathcal{G}$ among the multi-UAV system is undirected and connected, implying that there exists at least one valid path between any two nodes.

D. Convex Analysis and Prescribed-Time Stability

To guarantee the convergence and uniqueness of the solution to the distributed optimization problem, the following standard analytical assumption in multi-agent optimization is introduced.

Assumption 2: For all UAVs $i \in \mathcal{V}$, the local time-varying cost function $f_i(x_i, t) : \mathbb{R}^m \times \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}$ is twice continuously differentiable. Additionally, there exist positive constants $l \geq \mu > 0$ such that its Hessian matrix $H_i(x_i, t) = \nabla^2 f_i(x_i, t)$ satisfies the strong convexity and Lipschitz continuity conditions:

$$\mu I_m \leq H_i(x_i, t) \leq l I_m, \quad \forall x_i \in \mathbb{R}^m, t \geq 0$$

Furthermore, the partial derivative of the time-varying gradient $\frac{\partial \nabla f_i(x_i, t)}{\partial t}$ and its second-order partial derivative are globally bounded within their domains.

E. Useful Lemmas

Lemma 1 ([11]): For an undirected connected graph satisfying Assumption 1, its Laplacian matrix L is positive semi-definite, possessing a unique zero eigenvalue with $\mathbf{1}_N$ as the corresponding right eigenvector. Moreover, for any vector combination $x = [x_1^T, x_2^T, \dots, x_N^T]^T \in \mathbb{R}^{mN}$, the following algebraic connectivity eigenvalue inequality holds:

$$x^T (L \otimes I_m) x = \frac{1}{2} \sum_{i,j=1}^N a_{ij} \|x_i - x_j\|^2 \geq \lambda_2(L) \|x - \mathbf{1}_N \otimes \bar{x}\|^2$$

where $\lambda_2(L) > 0$ is the minimum non-zero eigenvalue (algebraic connectivity) of the Laplacian matrix, and $\bar{x} = \frac{1}{N} \sum_{i=1}^N x_i$ is the state mean.

Lemma 2 ([14]): Based on Assumption 2, the global total cost function $\sum_{i=1}^N f_i(x, t)$ is strongly convex. For an unconstrained distributed optimization problem, the cooperative state vector $x^*(t)$ satisfies the necessary and sufficient condition for its unique global minimum at time t if and only if:

$$\sum_{i=1}^N \nabla f_i(x^*(t), t) = \mathbf{0}_m$$

Lemma 3 ([27], [33]): Consider a continuous Lyapunov function $V(x, t) : \mathbb{R}^n \times \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$. If there exists a positive constant $k > 0$ such that the differential inequality holds along the system trajectories:

$$\dot{V}(x, t) \leq - \left(k + 2 \frac{\dot{\rho}(t; t_0, T)}{\rho(t; t_0, T)} \right) V(x, t)$$

where $\rho(t; t_0, T)$ is a smooth prescribed-time scaling function defined as:

$$\rho(t; t_0, T) = \begin{cases} \left(\frac{T}{t_0 + T - t} \right)^h, & t \in [t_0, t_0 + T) \\ 1, & t \geq t_0 + T \end{cases}$$

with a constant $h > 2$. Then, the system exhibits global prescribed-time stability, i.e., $\lim_{t \rightarrow (t_0+T)^-} V(x, t) = 0$, and for any $t \geq t_0 + T$, $V(x, t) \equiv 0$ holds.

III. MAIN RESULT

In this section, a fully distributed prescribed-time tracking control framework is developed for the second-order UAV swarm. The framework is mapped to the error dynamics, followed by a rigorous stability proof.

Consider the double-integrator model of the UAV system after feedback linearization: $\dot{p}_i = v_i$, $\dot{v}_i = u_i$. Defining the virtual cooperative state $x_i = p_i - \delta_i(t)$ yields the transformed state equation $\dot{x}_i = v_i - \dot{\delta}_i$. Introducing the virtual velocity v_{di} and defining the velocity tracking error as $\tilde{v}_i = v_i - v_{di} - \dot{\delta}_i$, the system is rewritten as the following error dynamics:

$$\dot{x}_i = \tilde{v}_i + v_{di} \quad (3a)$$

$$\dot{\tilde{v}}_i = u_i - \dot{v}_{di} - \ddot{\delta}_i \quad (3b)$$

A. Distributed Prescribed-Time Control Protocol

To achieve the coordinated optimization and collision avoidance, the virtual velocity v_{di} is designed as:

$$v_{di} = -\left(c_1 + \frac{\dot{\rho}(t)}{\rho(t)}\right) \left[\sum_{j \in \mathcal{N}_i} a_{ij} \alpha_{ij}(t) (x_i - x_j) + \Phi_i \right] \quad (4)$$

where $\alpha_{ij}(t)$ is the edge-based adaptive gain. To prevent meaningless accumulation after the prescribed time T , the update law adopts a piecewise locking design:

$$\dot{\alpha}_{ij} = \begin{cases} \Gamma_1 \rho(t) \dot{\rho}(t) \|x_i - x_j\|^2, & t \in [t_0, t_0 + T) \\ 0, & t \geq t_0 + T \end{cases} \quad (5)$$

with $\alpha_{ij}(t_0) = 0$ and learning rate $\Gamma_1 > 0$.

To address the inherent position-velocity coupling of the second-order dynamics and explicitly cancel the derivative of the virtual control law, the actual control law u_i is synthesized as:

$$u_i = -\beta_i(t) \left(c_2 + \frac{\dot{\rho}(t)}{\rho(t)} \right) \tilde{v}_i - \Phi_i, \quad t < t_0 + T \quad (6a)$$

$$- \sum_{j \in \mathcal{N}_i} a_{ij} (x_i - x_j) + \dot{v}_{di} + \ddot{\delta}_i$$

$$+ \xi_i - \text{sgn}(\tilde{v}_i) (\|\xi_i\|_\infty + \bar{D})$$

$$u_i = \xi_i + \ddot{\delta}_i - c_p \sum_{j \in \mathcal{N}_i} a_{ij} (x_i - x_j), \quad t \geq t_0 + T \quad (6b)$$

$$- c_d \sum_{j \in \mathcal{N}_i} a_{ij} (\dot{x}_i - \dot{x}_j)$$

where the node-based adaptive gain $\beta_i(t)$ is updated by:

$$\dot{\beta}_i = \begin{cases} \Gamma_2 \rho(t) \dot{\rho}(t) \|\tilde{v}_i\|^2, & t \in [t_0, t_0 + T) \\ 0, & t \geq t_0 + T \end{cases} \quad (7)$$

with $\beta_i(t_0) = 0$. Here, the parameter $\bar{D} \geq D_{\max}$ represents the upper bound of the lumped disturbances. The gradient compensation term ξ_i is defined as:

$$\xi_i = -H_i^{-1} \left(c_3 \nabla f_i + c_4 \left(H_i v_i + \frac{\partial \nabla f_i}{\partial t} \right) + \dot{H}_i v_i + \frac{\partial^2 \nabla f_i}{\partial t^2} \right) \quad (8)$$

where c_3 and c_4 are strictly positive constants.

The exact feedforward compensation term $\dot{v}_{di}$ in (6a) is expanded via the chain rule as:

$$\dot{v}_{di} = -\dot{\mu}(t) \left[\sum_{j \in \mathcal{N}_i} a_{ij} \alpha_{ij}(t) (x_i - x_j) + \Phi_i \right]$$

$$- \mu(t) \left[\sum_{j \in \mathcal{N}_i} a_{ij} \dot{\alpha}_{ij}(t) (x_i - x_j) \right]$$

$$- \mu(t) \left[\sum_{j \in \mathcal{N}_i} a_{ij} \alpha_{ij}(t) (v_i - \dot{\delta}_i - v_j + \dot{\delta}_j) + \dot{\Phi}_i \right]$$

where $\mu(t) = c_1 + \frac{\dot{\rho}(t)}{\rho(t)}$, $\dot{\mu}(t) = \frac{\ddot{\rho}(t)}{\rho(t)^2}$, and $\dot{\Phi}_i = \sum_{j \in \mathcal{N}_i} \nabla_{p_i}^2 U_{ij}(\|p_{ij}\|)(v_i - v_j)$.

Remark 1: Practical Considerations on Disturbance Bounds and Stability Degradation. While the robust switching term in the actual control law theoretically guarantees the suppression of lumped disturbances, it relies on a prior estimated upper bound $\bar{D}$. In highly complex real-world flight environments, such as sudden strong wind shears, strictly acquiring $\bar{D} \geq D_{\max}$ beforehand can be challenging. If the actual transient disturbance $d_i(t)$ exceeds the calibrated threshold $\bar{D}$, the residual uncompensated error $\tilde{v}_i^T(d_i(t) - \bar{D} \text{sgn}(\tilde{v}_i))$ cannot be completely canceled out in the Lyapunov derivative. In such extreme scenarios, the system's rigorous prescribed-time convergence will gracefully degrade: the UAV swarm will converge to a bounded residual set around the optimal trajectory rather than achieving zero steady-state error.

B. Feasibility of Collision Avoidance

The theoretical and practical feasibility of the proposed APF mechanism relies on two critical aspects:

1) *Validity of Initial Conditions:* The collision-avoidance guarantee requires a finite initial potential energy $V_1(t_0) < \infty$, necessitating $\|p_{ij}(t_0)\| > d_{\min}, \forall i \neq j \in \mathcal{V}$. This spatial constraint is naturally satisfied in practice, as UAVs launch from safely separated positions.

2) *Local Minima Circumvention:* Traditional APF methods often suffer from local minima where repulsive and attractive gradients cancel out, yielding $\Phi_i + \sum a_{ij} \alpha_{ij} (x_i - x_j) = \mathbf{0}$. Here, the time-varying formation offset $\delta_i(t)$ and target trajectory $x^*(t)$ act as continuous exogenous perturbations. Concurrently, the monotonically increasing adaptive gain, satisfying $\dot{\alpha}_{ij}(t) \geq 0$, reshapes the consensus attraction. These coupled mechanisms persistently break any static force equilibrium to ensure local minima circumvention.

Remark 2: Avoidance of Spatial Singularities in Backstepping. The feedforward compensation $\dot{v}_{di}$ requires the time derivative of the APF gradient, computed as $\dot{\Phi}_i = \sum_{j \in \mathcal{N}_i} \nabla_{p_i}^2 U_{ij}(\|p_{ij}\|)(v_i - v_j)$. Since the framework guarantees $\|p_{ij}(t)\| > d_{\min}$ for all $t \geq t_0$, the potential function U_{ij} remains infinitely differentiable. Consequently, the Hessian matrix $\nabla_{p_i}^2 U_{ij}$ is analytic and uniformly bounded, ensuring the high-order differentiation in backstepping avoids spatial singularities.

C. Properties and Parameter Tuning of Adaptive Gains

The proposed fully distributed adaptive gains possess the following properties, providing a theoretical foundation for parameter tuning:

1) *Monotonicity and Non-Singularity*: Based on (5) and (7), the gain derivatives are non-negative. Integrating over $t \in [t_0, t_0 + T)$ yields:

$$\alpha_{ij}(t) = \Gamma_1 \int_{t_0}^t \rho(\tau) \dot{\rho}(\tau) \|x_i(\tau) - x_j(\tau)\|^2 d\tau \geq 0$$

and similarly $\beta_i(t) \geq 0$. This monotonic growth from zero initial conditions, where $\alpha_{ij}(t_0) = \beta_i(t_0) = 0$, precludes zero-crossing singularities and avoids high-frequency chattering during transient convergence.

2) *Topological Robustness*: The spatial convergence margin depends on the dissipative term $4c_1\alpha^*\lambda_2(L)$ in (13). To sustain a convergence rate k under dynamic topologies, the steady-state gain must satisfy:

$$\alpha^* \geq \frac{k}{4c_1\lambda_2(L)}$$

If network deteriorations drop $\lambda_2(L)$, the transient spatial error $\|x_i - x_j\|$ inflates. This inflation accelerates the accumulation of $\alpha_{ij}(t)$, automatically driving the coupling strength to satisfy the new lower bound. This mechanism ensures resilience against topological reconfigurations without requiring global network knowledge.

3) *Transient Performance Tuning*: The edge gain α_{ij} acts as spatial structural stiffness, while the node gain β_i provides velocity damping. Their evolutionary rates, Γ_1 and Γ_2 , dictate the transient profile: increasing Γ_1 accelerates formation assembly but amplifies virtual velocity commands v_{di} , whereas increasing Γ_2 suppresses velocity tracking overshoot. Therefore, Γ_1 and Γ_2 must be balanced to optimize settling characteristics while adhering to the actuator saturation limit $\|u_i(t)\| \leq U_{\max}$.

D. Stability Analysis

Theorem 1: Under Assumptions 1 and 2, the multi-UAV system (3) driven by the fully distributed control laws (4) and (6) achieves the following objectives:

1) *The spatial consensus and velocity tracking errors satisfy:*

$$\lim_{t \rightarrow (t_0+T)^-} \|x_i(t) - x_j(t)\| = 0, \quad \lim_{t \rightarrow (t_0+T)^-} \|\tilde{v}_i(t)\| = 0$$

and the collision avoidance constraint $\|p_i(t) - p_j(t)\| > d_{\min}$ is strictly preserved for all $t \geq t_0$;

2) *The asymptotic optimal tracking is achieved for $t \geq t_0 + T$, satisfying:*

$$\lim_{t \rightarrow \infty} \sum_{i=1}^N \nabla f_i(x_i(t), t) = \mathbf{0}$$

3) *The actual control inputs remain globally bounded, satisfying:*

$$\limsup_{t \rightarrow (t_0+T)^-} \|u_i(t)\| < \infty$$

The detailed proof of Theorem 1 is provided in Appendix A.

Remark 3: Self-Adaptive Decoupling of Cross-Coupling Terms. The bounding process via Young's inequality highlights the necessity of the edge-based adaptive gain α_{ij} . Specifically, the adaptive law estimates a sufficiently large ideal gain α^* to ensure the negative-definite dissipative term dominates the positive-definite state coupling residuals. This mechanism resolves the non-collinear vector inner products induced by formation geometric offsets, decoupling complex nonlinearities inherent to second-order multi-agent systems.

Remark 4: Fully Distributed Property and Algebraic Connectivity. While the convergence margin k depends on the global algebraic connectivity $\lambda_2(L)$, practical deployment of the proposed protocol requires no prior topological knowledge. Physically, $\lambda_2(L)$ dictates the network's interaction strength; under weak connectivity with small $\lambda_2(L)$, a larger ideal gain α^* is necessary for the dissipative term $4c_1\alpha^*\lambda_2(L)$ to suppress cross-couplings and disturbances. The edge-based gain $\alpha_{ij}(t)$ dynamically adapts to a coupling threshold commensurate with the underlying topology, underscoring the plug-and-play advantage of the fully distributed architecture.

Remark 5: Selection Criterion for Time-Scaling Parameter h . Unlike first-order systems requiring $h > 1$ for bounded inputs, the second-order backstepping design introduces high-order feedforward compensation involving $\dot{\mu}(t)$. This necessitates $h > 2$ to offset the second-order singularity. Specifically, selecting $h = 2.1$ prevents numerical explosion and ensures a smooth transition from the prescribed-time rendezvous to the asymptotic optimal tracking.

The two-phase framework features a piecewise convergence mechanism. For $t < t_0 + T$, the prescribed-time scaling function forces precise spatial configuration rendezvous. Upon entering the tracking phase at $t \geq t_0 + T$, the system transitions to a linear time-invariant error dynamic, achieving asymptotically stable tracking of the time-varying global optimal trajectory. This composite architecture bypasses the control singularities and high-frequency chattering of traditional finite/fixed-time algorithms, generating physically realizable actuator commands.

IV. NUMERICAL SIMULATIONS AND ANALYSIS

In this section, the proposed fully distributed prescribed-time optimization framework is systematically verified. To ensure theoretical rigor and avoid coincidental conclusions from single-run tests, the experiments are conducted progressively: starting from basic scenario validation and ablation studies, advancing to complex extensible scenarios, performing quantitative comparative analyses, and concluding with Monte Carlo statistical evaluations and robustness tests against communication delays.

A. Experimental Setup and Quantitative Metrics

Consider a swarm of $N = 5$ UAVs assigned to track a moving target in 3-D space. The communication topology $\mathcal{G}$ adopts an undirected fully connected structure. To comprehensively evaluate the algorithm's robustness, the initial positions $p_i(0)$ are subjected to random perturbations in subsequent statistical

TABLE II. Initial conditions of UAVs and target

	Position [m]	Velocity [m/s]
UAV 1	$[-67.6, -76.5, 200]^T$	$[25, 25, 0]^T$
UAV 2	$[-67.6, -123.5, 200]^T$	$[25, 25, 0]^T$
UAV 3	$[-112.4, -138, 200]^T$	$[25, 25, 0]^T$
UAV 4	$[-140, -100, 200]^T$	$[25, 25, 0]^T$
UAV 5	$[-112.4, -62, 200]^T$	$[25, 25, 0]^T$
Target	$[0, 0, 0]^T$	$[8, 15, 0]^T$

tests, while the nominal initial states of the UAVs and the target are detailed in Table II.

The basic control parameters are configured as: prescribed time $T = 15$ s, time-scaling exponent $h = 2.1$, feedback gains $c_1 = 0.0, c_2 = 0.5, c_3 = 0.1, c_4 = 0.8$, and learning rates $\Gamma_1 = \Gamma_2 = 10^{-3}$. To prevent chattering, a smooth hyperbolic tangent function $\tanh(10\tilde{v}_i)$ is utilized, and the actuator saturation limit is strictly enforced at $U_{\max} = 15$ m/s².

Moving beyond visual assessments, the following core metrics are mathematically defined to provide a rigorous quantitative evaluation and statistical analysis across 20 independent Monte Carlo runs:

- **Prescribed-Time Consensus Error $E_{form}(T)$:** Evaluates the exact spatial formation assembly precision at $t = T$, defined as $E_{form}(T) = \max_i \|x_i(T) - \bar{x}(T)\|$.
- **Tracking RMSE:** Root Mean Square Error evaluating the steady-state tracking precision, defined as $RMSE = \sqrt{\frac{1}{t_{end}-T} \int_T^{t_{end}} \frac{1}{N} \sum_{i=1}^N \|x_i(\tau) - x^*(\tau)\|^2 d\tau}$.
- **Control Energy E_u :** Evaluates the actuator effort and energy economy, defined as $E_u = \int_0^{t_{end}} \sum_{i=1}^N \|u_i(\tau)\|^2 d\tau$.
- **Safety Distance Margin D_{safe} :** Ensures collision avoidance, defined as $D_{safe} = \min_{t,i \neq j} \|p_i(t) - p_j(t)\| - d_{min}$. A valid APF guarantees $D_{safe} > 0$.

Remark 6: Engineering Considerations on Initial Feasibility and Disturbance Bounds. For real-world UAV deployments, the upper bound of lumped disturbances $\bar{D}$ can be conservatively calibrated offline via wind tunnel tests or motor static thrust limits. If $\bar{D}$ cannot be accurately estimated, the system gracefully degrades from zero steady-state error to bounded convergence. Furthermore, to guarantee the initial feasibility of the APF mechanism, a pre-processing protocol is embedded: if extreme initial conditions yield $\|p_{ij}(0)\| \leq d_{min}$, the system briefly activates only the repulsive force to safely scatter the UAVs before initializing the prescribed-time main controller.

B. Basic Scenario Validation

This basic scenario validates the prescribed-time tracking performance and numerical stability. The target trajectory is a long-range S-shaped curve: $p_{rx} = 2t$ and $p_{ry} = 40 \sin(0.02\pi t)$.

As illustrated in Fig. 2, during Phase 1 ($t < 15$ s), all UAVs rapidly rendezvous toward the target center. The integrated APF mechanism maintains safe inter-UAV distances, allowing the swarm to form the pentagram and track the target exactly at the prescribed deadline $T = 15$ s.

Driven by the prescribed-time scaling function, the swarm prioritizes precise formation assembly within the first 15 sec-

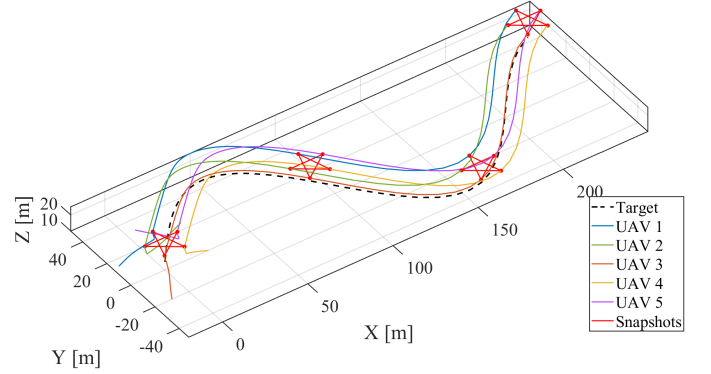


Fig. 2: 3-D S-shaped cooperative tracking trajectories.

onds (Figs. 3 and 4). After switching to Phase 2 at $T = 15$ s, the velocity error exhibits a gentle catch-up transient and settles to zero. The overall tracking RMSE is 11.0430 m (dominated by the Phase 1 transient lag), with a total control energy of 3511.0292 (m/s²)²·s.

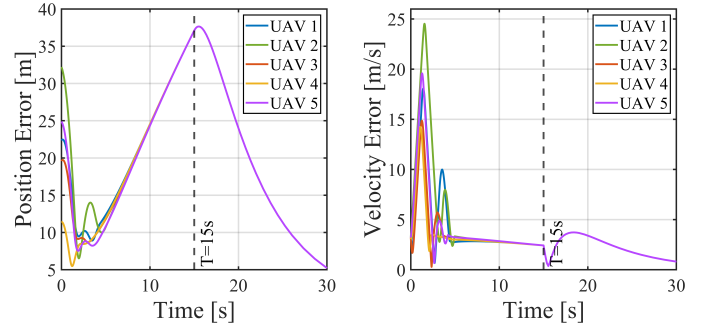


Fig. 3: Position and velocity tracking errors.

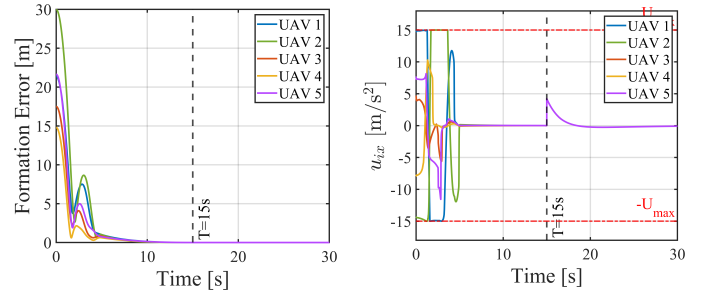
Fig. 4: Formation consensus error $\|x_i - \bar{x}\|$.

Fig. 5: Controller inputs during tracking.

Prior to 15 s, the adaptive gains rapidly increase, as detailed in Fig. 6 and Table III. Once $t \geq 15$ s, all parameters are locked. This piecewise locking mechanism prevents infinite gain inflation or switching jumps at $t = T$, enabling the control inputs to smoothly stabilize within saturation limits (Fig. 5).

C. Ablation Studies

Three dedicated ablation studies verify the core mechanisms of the proposed two-phase framework. The basic parameters remain identical to the nominal scenario, with specific modules enabled or disabled for comparison.

TABLE III. Numerical statistics of adaptive parameter evolution before and after the prescribed time $T = 15$ s

Time t [s]	Edge Gain α_{12}	Node Gain β_1	Node Gain β_2	Node Gain β_3	Node Gain β_4	Node Gain β_5
0.00	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
1.01	0.3565	0.0091	0.0065	0.0110	0.0120	0.0104
3.00	0.5209	0.0913	0.1220	0.0608	0.0389	0.0676
4.99	0.5984	0.1082	0.1597	0.0617	0.0389	0.0797
9.98	0.6052	0.1082	0.1597	0.0617	0.0389	0.0797
15.00	0.6054	0.1082	0.1597	0.0617	0.0389	0.0797
20.21	0.6054	0.1082	0.1597	0.0617	0.0389	0.0797
State	Locked	Locked	Locked	Locked	Locked	Locked

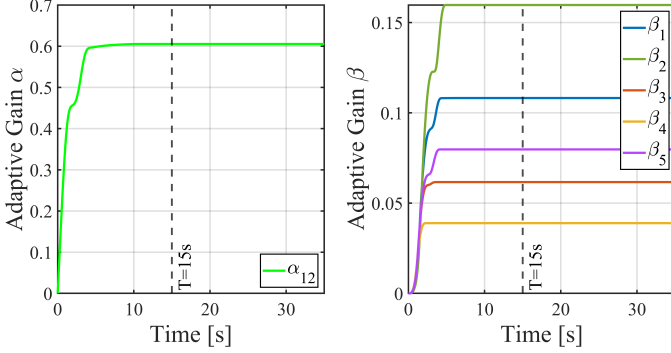


Fig. 6: Distributed adaptive gains.

1) *Ablation Study on Collision Avoidance:* The Artificial Potential Field mechanism is evaluated by comparing the minimum inter-UAV distances when enabling and disabling the avoidance parameters, specifically setting $d_{\min} = 0$ and $R_{apf} = 0$.

As shown in Fig. 7(a), the proposed method constrains the inter-UAV distance above the 1.0 m safety threshold during the highly dynamic transient phase. Conversely, without the APF mechanism detailed in Fig. 7(b), the minimum distance drops to 0.00 m at $t \approx 8$ s, causing a fatal collision. This confirms that the local repulsive force is an indispensable safety barrier.

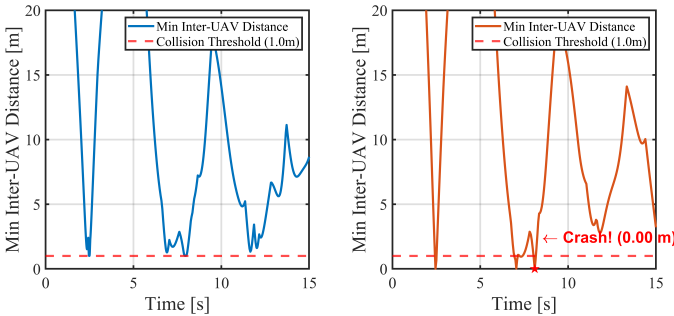


Fig. 7: Minimum inter-UAV distance with and without APF.

2) *Ablation Study on Fully Distributed Adaptive Gains:* Traditional prescribed-time protocols rely heavily on global topological information, such as the minimum non-zero eigenvalue λ_2 . To demonstrate how the proposed adaptive gains eliminate this reliance, an ablation study uses a weakly connected line topology bridging nodes 1 through 5.

Fig. 8 compares the formation consensus error $\|x_i - \bar{x}\|$. When conservative fixed gains are used without global knowl-

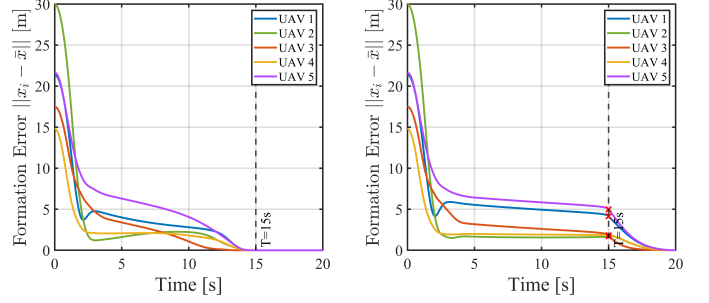


Fig. 8: Formation consensus error under a weak line topology: (a) Proposed adaptive gains; (b) Fixed gains.

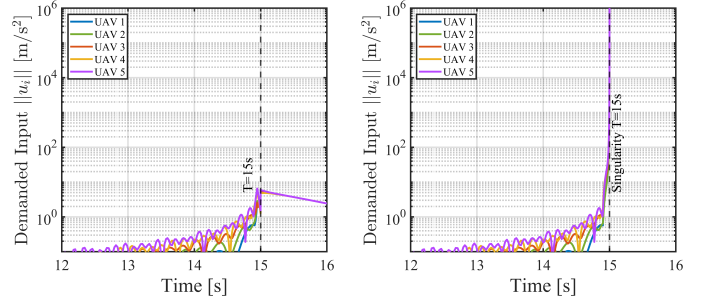


Fig. 9: Control inputs with and without state-switching.

edge, the control effort fails to overcome the weak connectivity, leaving a large residual error at $T = 15$ s as depicted in Fig. 8(b). In contrast, the proposed adaptive algorithm dynamically adjusts gains based on local interaction errors, as shown in Fig. 8(a). This accurately compensates for the weak topology, driving the error to zero at $T = 15$ s and guaranteeing plug-and-play robustness.

3) *Ablation Study on State-Switching Mechanism for Singularity Avoidance:* A critical challenge in prescribed-time control is the internal time singularity as $t \rightarrow T$. To illustrate the destabilizing impact of this temporal singularity, a microscopic sensor noise with a 0.005 m amplitude is injected, and the demanded control input $\|u_i\|$ is analyzed logarithmically.

As shown in Fig. 9(b), without the state-switching mechanism, infinite gain amplification near the singularity causes the minor noise to explode. The demanded acceleration exhibits a vertical asymptote, exceeding 10^5 m/s² at $t \approx 14.99$ s, which would trigger severe actuator saturation and high-frequency chattering. Under the proposed two-phase framework, the system switches to the singularity-free Phase 2 at $T = 15$ s, as illustrated in Fig. 9(a). This transition truncates the noise

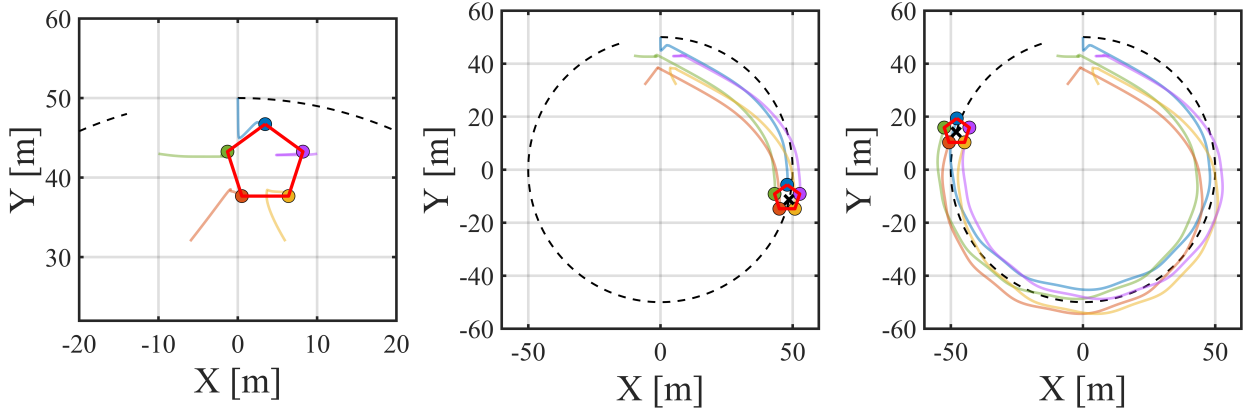


Fig. 10: Feature snapshots of the circular cooperative tracking: (a) Formation rendezvous completed at 15 s, (b) Dynamic catch-up at 36 s, (c) Precise tracking at 100 s.

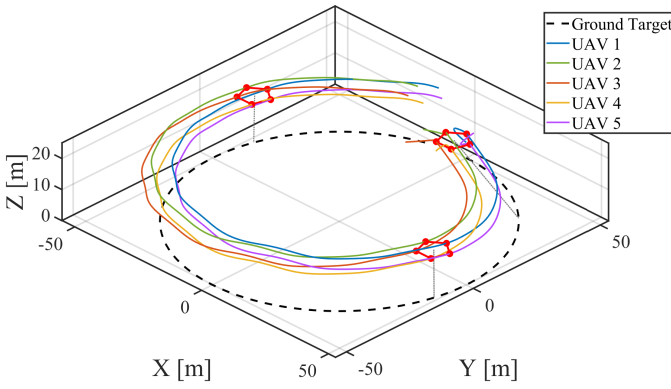


Fig. 11: 3-D trajectories for circular tracking.

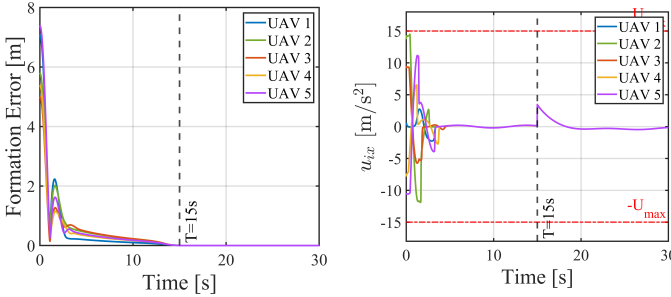


Fig. 12: Formation consensus error for circular tracking.

Fig. 13: Control inputs for circular tracking.

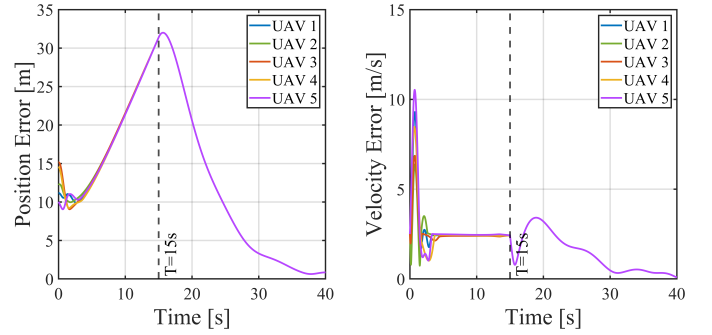


Fig. 14: Tracking errors for circular tracking.

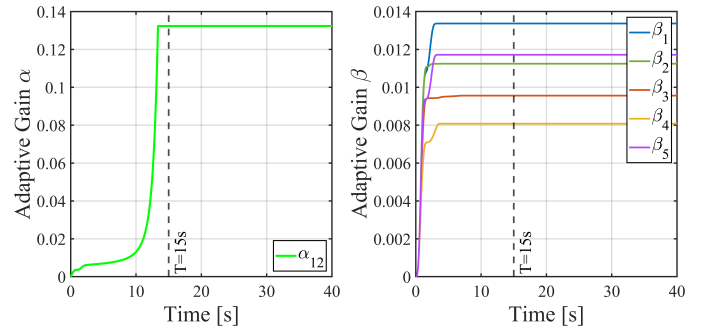


Fig. 15: Evolution trajectories of the fully distributed adaptive gains under the circular scenario.

amplification, allowing the control input to stabilize smoothly.

4) *Ablation Study on Time-Scaling Parameter h* : While Theorem 1 establishes $h > 2$ to prevent second-order control singularities, the specific selection of $h = 2.1$ requires engineering justification. To evaluate the impact of h , a statistical ablation study comprising 20 Monte Carlo runs with ± 20 m initial spatial perturbations is conducted under two scenarios: an ideal unconstrained environment with $U_{max} = \infty$ and a saturated environment limited to $U_{max} = 15$ m/s².

As demonstrated in Table IV, under ideal unconstrained dynamics, a larger h forces a steep convergence schedule. While this mathematical precision reduces the theoretical error to the 10^{-8} m scale, it induces an exponential surge in the peak

demand input exceeding 774 m/s² and control energy.

In physical deployments, as presented in Table V, such demanded peaks are unrealizable. Consequently, systems with larger h values, such as $h = 4.0$ and 5.0 , suffer from prolonged actuator saturation exceeding 82% to 89% of the mission duration. This continuous saturation breaks the theoretical time-scaling schedule. As $t \rightarrow T$, the unrecoverable spatial lag coupled with infinite gains causes the system to collapse, degrading the actual terminal formation error to the 4 to 9 meter scale. The proposed selection of $h = 2.1$ serves as an optimal engineering trade-off: it circumvents the second-order singularity while keeping trajectory demands aligned with physical actuator capabilities, guaranteeing 10^{-7} m terminal

TABLE IV. Ablation Study of h Under Ideal Dynamics Expressed as Mean $\pm$ Standard Deviation

h	U_{peak} [m/s ²]	E_u	E_{form} [m]
2.1	229.35 $\pm$ 46.19	30488.13 $\pm$ 12121.99	$6.04 \times 10^{-7} \pm 4.01 \times 10^{-7}$
3.0	239.23 $\pm$ 56.25	32096.24 $\pm$ 13384.60	$2.46 \times 10^{-7} \pm 1.95 \times 10^{-7}$
4.0	262.49 $\pm$ 68.55	35495.06 $\pm$ 15857.43	$1.30 \times 10^{-7} \pm 1.63 \times 10^{-7}$
5.0	774.85 $\pm$ 2172.97	90502.22 $\pm$ 229420.56	$7.19 \times 10^{-8} \pm 1.03 \times 10^{-7}$

TABLE V. Ablation Study of h Under Saturation Limit $U_{max} = 15$ m/s² Expressed as Mean $\pm$ Standard Deviation

h	Sat. Ratio [%]	E_u	E_{form} [m]
2.1	65.60 $\pm$ 6.70	4214.00 $\pm$ 688.11	$4.35 \times 10^{-7} \pm 4.46 \times 10^{-7}$
3.0	70.59 $\pm$ 11.29	4725.03 $\pm$ 2322.43	0.21 $\pm$ 0.68
4.0	82.70 $\pm$ 16.21	8186.06 $\pm$ 5754.18	4.14 $\pm$ 6.57
5.0	89.37 $\pm$ 14.35	11562.35 $\pm$ 6644.84	9.19 $\pm$ 10.54

precision even under severe initial deviations.

D. Algorithm Extensibility Verification: Complex Circular Maneuver Tracking

Expanding beyond nominal conditions, a ground-relative circular maneuvering scenario verifies the algorithm's robustness and the "rendezvous first, track later" characteristic in highly dynamic tasks.

Fig. 11 and the dynamic feature snapshots in Fig. 10 illustrate the spatial evolution of the swarm. The simulation trajectories corroborate the temporal priority of the control logic: during Phase 1, $t < 15$ s, the swarm prioritizes formation reconfiguration, accomplishing the collision-free rendezvous of the pentagram configuration, as shown in Fig. 10(a). Subsequently, for $t \geq 15$ s, the formation with locked adaptive gains smoothly converges to and tracks the target's circular trajectory, depicted in Figs. 10(b) and (c).

The error evolution curves in Figs. 12 and 14 provide a numerical basis for the physical process. As shown in Fig. 12, driven by the time-scaling function, the formation consensus error converges to the origin exactly at $T = 15$ s. Since most control energy in Phase 1 is devoted to satisfying the prescribed-time consensus constraint, the position tracking error in Fig. 14 exhibits a transient lag, peaking at 15 s. After transitioning into Phase 2, the asymptotic stability mechanism takes over, eliminating the tracking deviation and achieving dynamic tracking without steady-state errors around 40 s.

Additionally, the smooth acceleration input responses in Fig. 13 demonstrate that the piecewise locking and smoothing mechanisms avoid numerical singularities and abrupt spikes even during the phase-switching catch-up period near $t = 15$ s. The inputs strictly obey the actuator saturation limit of $U_{max} = 15$ m/s². This validates the control energy economy and engineering feasibility of the proposed algorithm in highly dynamic scenarios.

Lastly, the adaptive parameters in Fig. 15 adhere to the theoretical derivation. The gains α and β smoothly increase driven by interaction errors during the first 15 s. Upon reaching $T = 15$ s, the locking mechanism is triggered, and all gains are clamped to constants. This prevents numerical divergence at

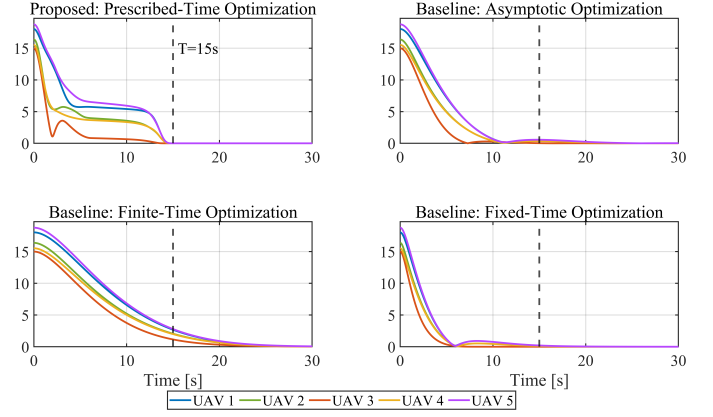


Fig. 16: Consensus error comparison among algorithms.

the temporal singularity, corroborating the robustness of the parameter locking mechanism.

E. Comparative and Statistical Analysis

A comparative analysis against Asymptotic, Finite-Time, and Fixed-Time distributed consensus algorithms benchmarks the proposed framework. Furthermore, 20 independent Monte Carlo runs with uniform initial spatial perturbations of ± 20 m and velocity perturbations of ± 5 m/s evaluate statistical robustness. Performance is assessed across four metrics: Prescribed-Time Formation Error $E_{form}(T)$, Steady-State Tracking RMSE, Total Control Energy E_u , and Peak Control Input U_{peak} , with results summarized in Table VI.

As demonstrated in Table VI and Fig. 16, the proposed control achieves a terminal formation error of 8.16×10^{-6} m at $T = 15$ s. Conversely, the Asymptotic and Finite-Time algorithms stall at the 10^{-2} m magnitude under severe initial deviations. This indicates that traditional time-constrained protocols fail to fulfill temporal mission requirements under substantial spatial perturbations. Moreover, the 3.12×10^{-6} m standard deviation of the proposed method's terminal error verifies its robustness against external uncertainties.

The proposed framework excels in physical feasibility. While the Fixed-Time algorithm reaches 3.34×10^{-5} m accuracy, it incurs a high control cost. As shown in Fig. 17, the peak control input of the Fixed-Time algorithm surges to 210.70 ± 66.82 m/s², violating the typical 15 m/s² saturation limit of physical UAV actuators. In contrast, the proposed algorithm avoids control singularity, bounding the peak input near 16.61 ± 3.41 m/s². This ensures rapid spatial convergence without inducing actuator breakdown.

Regarding the total control energy E_u , the low consumption of the Asymptotic and Finite-Time algorithms is a consequence of their inadequate control effort, leading to mission failure. Among algorithms forcing rapid convergence, the proposed method distributes control energy more efficiently with 423.32 units compared to the Fixed-Time approach's 576.22 units. Furthermore, the proposed algorithm yields a tracking RMSE of 2.01 ± 0.15 m, marginally higher than the Fixed-Time algorithm's 1.39 ± 0.11 m. This reflects a deliberate design trade-off. Rather than demanding aggressive high-gain corrections instantly, the two-phase mechanism

TABLE VI. Statistical Performance Comparison Over 20 Monte Carlo Runs (Mean $\pm$ Std. Dev.)

Algorithm	Requires $\lambda_2(L)$?	$E_{form}(T = 15s)$ [m]	RMSE [m]	E_u [(m/s) ² ·s]	U_{peak} [m/s ²]
Proposed (Ours)	No (Plug-and-play)	$8.16 \times 10^{-6} \pm 3.12 \times 10^{-6}$	2.01 ± 0.15	423.32 ± 92.10	16.61 ± 3.41
Asymptotic	No	$1.46 \times 10^{-2} \pm 1.38 \times 10^{-3}$	2.05 ± 0.14	62.43 ± 11.29	8.71 ± 1.75
Finite-Time	Yes	$1.25 \times 10^{-2} \pm 9.83 \times 10^{-4}$	2.50 ± 0.19	24.68 ± 2.38	3.31 ± 0.42
Fixed-Time	Yes	$3.34 \times 10^{-5} \pm 7.19 \times 10^{-6}$	1.39 ± 0.11	576.22 ± 62.32	210.70 ± 66.82

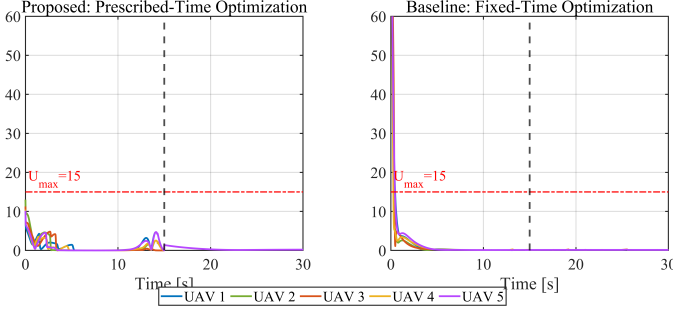


Fig. 17: Comparison of demanded control inputs between the proposed method and the Fixed-Time optimization.

Algorithm 1: Implementation of the Two-Phase PTC Framework

Input: $T, \Delta t, \lambda = 0.95, U_{max}$
1 Init: $t = 0, \alpha_{ij} = 0, \beta_i = 0, \mathbf{u}_{ic} = \mathbf{0}$;
2 while *Mission active* **do**
 3 Receive states and compute $\mathbf{x}_i[k]$ and $\Phi_i[k]$;
 4 if $t < T$ **then**
 5 Update adaptive gains α_{ij}, β_i via (14) and (16);
 6 Compute raw control $\mathbf{u}_{raw}[k]$ via Phase 1 protocol (15a);
 7 else
 8 Lock α_{ij}, β_i and compute $\mathbf{u}_{raw}[k]$ via Phase 2 protocol (15b);
 9 end
 10 $\mathbf{u}_{ic}[k] \leftarrow \lambda \mathbf{u}_{ic}[k-1] + (1-\lambda) \mathbf{u}_{raw}[k]$;
 11 Apply $\mathbf{u}_i[k] = \text{sat}(\mathbf{u}_{ic}[k], U_{max})$ and $t \leftarrow t + \Delta t$;
12 end

ROS 2 Control Node (FiveUAVCircleTracker)

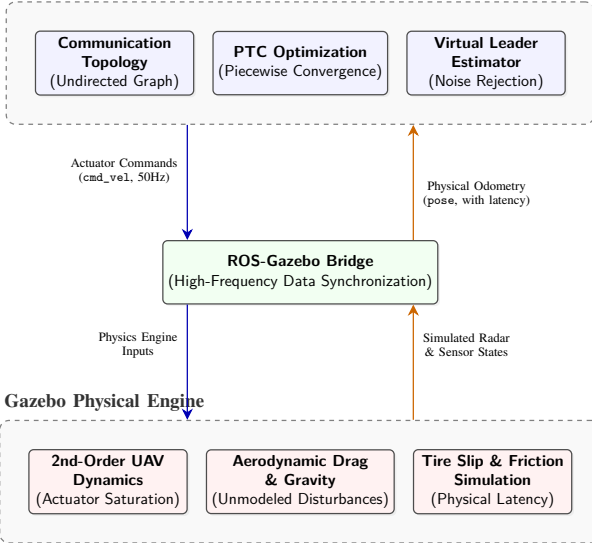


Fig. 18: System architecture of the Hardware-in-the-Loop (HIL) validation environment.

maintains gentle adjustments during the initial transient phase to prevent actuator saturation and high-frequency chattering. By concentrating control authority as time approaches 15 s, the algorithm exchanges a minor increase in average error for terminal convergence and hardware safety, underscoring its engineering practicality.

V. HARDWARE-IN-THE-LOOP VALIDATION

Bridging the gap between theoretical derivation and physical implementation, a Hardware-in-the-Loop (HIL) simulation is conducted using ROS 2 and Gazebo to evaluate the framework's resilience against real-world aerodynamic disturbances.

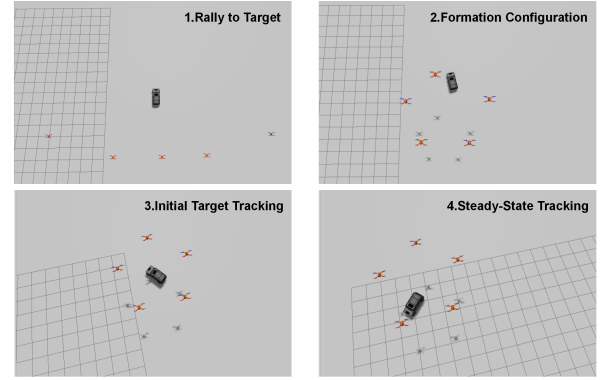


Fig. 19: Feature snapshots of the cooperative tracking in the Gazebo environment.

A. Platform Setup and Scenario

Unlike idealized numerical simulations, the HIL environment introduces unmodeled aerodynamic drag, physical latency, and discrete sampling effects. As illustrated in Fig. 18, the proposed algorithm is deployed as a ROS 2 control node, communicating with five full-physics quadrotors and a ground vehicle via the `ros_gz_bridge` at 50 Hz. To mitigate latency-induced desynchronization, an odometry-based virtual leader estimator is incorporated.

In the test scenario, the ground target travels along a 60 m radius circular trajectory at 3 m/s. The 5-UAV swarm is tasked to assemble a regular pentagram formation, with a 10 m radius and 15 m altitude, above the moving target within the prescribed time $T = 15$ s. Actuator saturation is enforced at 15 m/s².

B. Tracking Results and Engineering Trade-offs

Fig. 19 depicts the spatial evolution of the swarm. During the transient rendezvous phase, $t < 15$ s, the UAVs converge

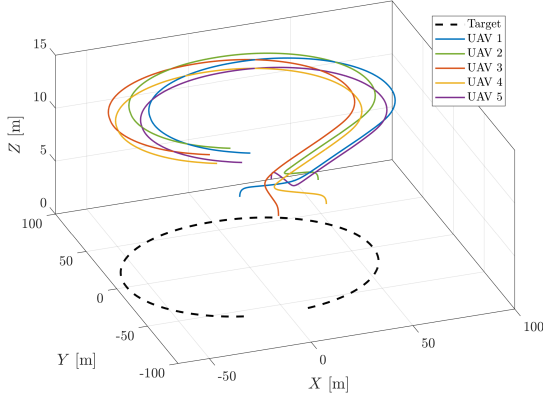


Fig. 20: 3-D tracking trajectories in the HIL simulation.

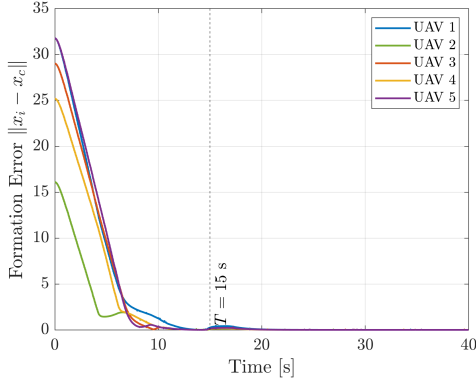


Fig. 21: Formation consensus error in HIL simulation.

while the APF mechanism guarantees collision avoidance. At $T = 15$ s, the formation is established, as shown in Fig. 20. Validated by the consensus error plot in Fig. 21, the formation error converges to zero at 15 s and remains locked.

Notably, a bounded steady-state tracking error emerges after $t = 15$ s, depicted in Fig. 22. This reflects a deliberate engineering trade-off. While the theoretical derivation guarantees zero steady-state error, the digital implementation of high-gain prescribed-time controllers near the singularity $t \rightarrow T$ faces discrete-time integration divergence at 50 Hz.

To ensure numerical robustness, a digital low-pass velocity filter with a 0.95 damping factor is embedded to emulate physical aerodynamic drag. Forcing zero tracking error at the singularity would trigger actuator saturation and potential hardware breakdown. Consequently, the framework implements a "graceful degradation" strategy. It allows a minor trajectory lag to serve as a numerical damping buffer, neutralizing high-frequency chattering while prioritizing formation integrity.

VI. CONCLUSION

This article proposes a fully distributed prescribed-time tracking framework for second-order UAV swarms. Within this framework, an adaptive gain mechanism eliminates the reliance on global Laplacian eigenvalues, while a backstepping controller analytically decouples position-velocity dynamics to bypass inherent infinite-gain singularities. Additionally, a

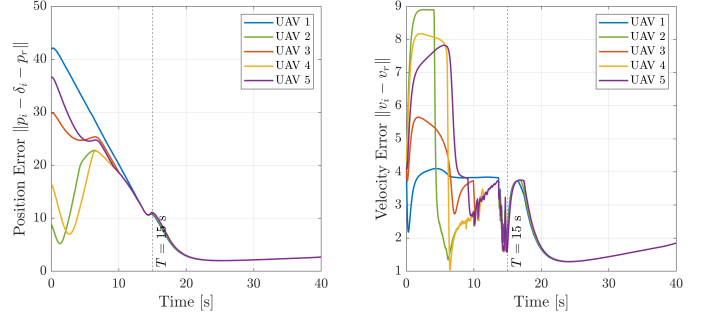


Fig. 22: Position and velocity tracking errors in HIL simulation.

state-switching protocol guarantees collision-free formation assembly and subsequent asymptotic optimal tracking. Numerical and HIL simulations validate the method's robust convergence, computational economy, and practical feasibility.

APPENDIX A PROOF OF THEOREM 1

Proof: The stability analysis is conducted in the following three steps.

Step 1: Prescribed-Time Consensus and Collision Avoidance ($t < t_0 + T$).

Consider the candidate Lyapunov function:

$$V_1(t) = \frac{1}{4} \sum_{i=1}^N \sum_{j=1}^N a_{ij} \|x_i - x_j\|^2 + \frac{1}{2} \sum_{i=1}^N \|\tilde{v}_i\|^2 + \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N U_{ij}(\|p_{ij}\|) + \sum_{i=1}^N \sum_{j=1}^N \frac{(\alpha_{ij} - \alpha^*)^2}{2\Gamma_1 \rho(t)^2} + \sum_{i=1}^N \frac{(\beta_i - \beta^*)^2}{2\Gamma_2 \rho(t)^2}$$

The five terms in $V_1(t)$ respectively capture: (i) spatial consensus error, (ii) velocity tracking error, (iii) collision avoidance potential energy, (iv) edge-gain estimation error, and (v) node-gain estimation error. Here, α^* and β^* are unknown positive constants. Taking the time derivative of $V_1(t)$ and utilizing $\frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N \dot{U}_{ij} = \sum_{i=1}^N \Phi_i^T \dot{p}_i$ gives:

$$\begin{aligned} \dot{V}_1 = & \sum_{i=1}^N \tilde{v}_i^T \left[\sum_{j=1}^N a_{ij} (x_i - x_j) + \Phi_i + u_i + d_i(t) - \dot{v}_{di} - \ddot{\delta}_i \right] \\ & + \sum_{i=1}^N \left(\sum_{j=1}^N a_{ij} (x_i - x_j) + \Phi_i \right)^T v_{di} - 2 \frac{\dot{\rho}}{\rho} (V_\alpha + V_\beta) \\ & + \sum_{i=1}^N \sum_{j=1}^N \frac{(\alpha_{ij} - \alpha^*) \dot{\alpha}_{ij}}{\Gamma_1 \rho^2} + \sum_{i=1}^N \frac{(\beta_i - \beta^*) \dot{\beta}_i}{\Gamma_2 \rho^2} \end{aligned} \quad (9)$$

Substituting (6a) into (9) yields:

$$\begin{aligned} & \sum_{i=1}^N \tilde{v}_i^T (\xi_i + d_i(t) - \text{sgn}(\tilde{v}_i)(\|\xi_i\|_\infty + \bar{D})) \\ & \leq \sum_{i=1}^N \left(\|\tilde{v}_i\|_1 \|\xi_i\|_\infty + \|\tilde{v}_i\|_1 \bar{D} - \|\tilde{v}_i\|_1 (\|\xi_i\|_\infty + \bar{D}) \right) = 0 \end{aligned}$$

Thus, one has:

$$\begin{aligned} \dot{V}_1 \leq & \sum_{i=1}^N \left(\sum_{j=1}^N a_{ij} (x_i - x_j) + \Phi_i \right)^T v_{di} \\ & - \sum_{i=1}^N \beta_i \left(c_2 + \frac{\dot{\rho}}{\rho} \right) \|\tilde{v}_i\|^2 - 2\frac{\dot{\rho}}{\rho} (V_\alpha + V_\beta) \\ & + \sum_{i=1}^N \sum_{j=1}^N \frac{(\alpha_{ij} - \alpha^*) \dot{\alpha}_{ij}}{\Gamma_1 \rho^2} + \sum_{i=1}^N \frac{(\beta_i - \beta^*) \dot{\beta}_i}{\Gamma_2 \rho^2} \end{aligned} \quad (10)$$

Substituting (4) into (10) yields:

$$\begin{aligned} \dot{V}_1 \leq & - \sum_{i=1}^N \left(c_1 + \frac{\dot{\rho}}{\rho} \right) \left(\sum_{j=1}^N a_{ij} (x_i - x_j) \right)^T \\ & \times \left(\sum_{j=1}^N a_{ij} \alpha_{ij} (x_i - x_j) \right) - \sum_{i=1}^N \left(c_1 + \frac{\dot{\rho}}{\rho} \right) \|\Phi_i\|^2 \\ & - \sum_{i=1}^N \left(c_1 + \frac{\dot{\rho}}{\rho} \right) \Phi_i^T \sum_{j=1}^N a_{ij} (1 + \alpha_{ij}) (x_i - x_j) \\ & - \sum_{i=1}^N \beta_i \left(c_2 + \frac{\dot{\rho}}{\rho} \right) \|\tilde{v}_i\|^2 - 2\frac{\dot{\rho}}{\rho} (V_\alpha + V_\beta) \\ & + \sum_{i=1}^N \sum_{j=1}^N \frac{(\alpha_{ij} - \alpha^*) \dot{\alpha}_{ij}}{\Gamma_1 \rho^2} + \sum_{i=1}^N \frac{(\beta_i - \beta^*) \dot{\beta}_i}{\Gamma_2 \rho^2} \end{aligned} \quad (11)$$

Applying Young's inequality to the cross-coupling term gives:

$$\begin{aligned} & - \sum_{i=1}^N \left(c_1 + \frac{\dot{\rho}}{\rho} \right) \Phi_i^T \sum_{j=1}^N a_{ij} (1 + \alpha_{ij}) (x_i - x_j) \\ & \leq \sum_{i=1}^N \left(c_1 + \frac{\dot{\rho}}{\rho} \right) \left[\frac{1}{2} \|\Phi_i\|^2 + \frac{1}{2} \left\| \sum_{j=1}^N a_{ij} (1 + \alpha_{ij}) (x_i - x_j) \right\|^2 \right] \end{aligned}$$

Defining $\tilde{\alpha}_{ij} = \alpha_{ij} - \alpha^*$ and $\tilde{\beta}_i = \beta_i - \beta^*$, and substituting (5) and (7) into (11) yields:

$$\begin{aligned} \dot{V}_1 \leq & - \sum_{i=1}^N \sum_{j=1}^N a_{ij} \alpha^* \left(c_1 + \frac{\dot{\rho}}{\rho} \right) \|x_i - x_j\|^2 - \sum_{i=1}^N \beta^* \left(c_2 + \frac{\dot{\rho}}{\rho} \right) \|\tilde{v}_i\|^2 \\ & - \frac{1}{2} \sum_{i=1}^N \left(c_1 + \frac{\dot{\rho}}{\rho} \right) \|\Phi_i\|^2 - 2\frac{\dot{\rho}}{\rho} (V_\alpha + V_\beta) \end{aligned} \quad (12)$$

Using Lemma 1, $\frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N a_{ij} \|x_i - x_j\|^2 \geq \lambda_2(L) \|x - \mathbf{1}_N \otimes \bar{x}\|^2$. Defining $k = \min(4c_1\alpha^*\lambda_2(L), 2c_2\beta^*) > 0$, (12) is bounded as:

$$\dot{V}_1(t) \leq -k(V_x + V_v) - 2\frac{\dot{\rho}(t)}{\rho(t)} V_1(t) \leq -\left(k + 2\frac{\dot{\rho}(t)}{\rho(t)}\right) V_1(t) \quad (13)$$

By Lemma 3, $V_1(t) \rightarrow 0$ as $t \rightarrow (t_0 + T)^-$, ensuring $\lim_{t \rightarrow (t_0+T)^-} \|x_i(t) - x_j(t)\| = 0$ and $\lim_{t \rightarrow (t_0+T)^-} \|\tilde{v}_i(t)\| = 0$. Since $V_1(t) \leq V_1(t_0)$, $\frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N U_{ij}(\|p_{ij}(t)\|) \leq V_1(t_0)$ holds, ensuring $\|p_i(t) - p_j(t)\| > d_{\min}$ for all $t \in [t_0, t_0 + T)$.

Step 2: Asymptotic Optimal Tracking ($t \geq t_0 + T$).

For $t \geq t_0 + T$, $x_i = x_j$ and $\Phi_i \equiv \mathbf{0}$. Substituting (6b) into $\ddot{x}_i = u_i - \ddot{\delta}_i$ and left-multiplying by H_i yields:

$$H_i \ddot{x}_i = H_i \xi_i - c_p \sum_{j \in \mathcal{N}_i} a_{ij} (x_i - x_j) - c_d \sum_{j \in \mathcal{N}_i} a_{ij} (\dot{x}_i - \dot{x}_j)$$

Summing the above equation over $i = 1, \dots, N$ and utilizing the Laplacian zero-row-sum property $\mathbf{1}_N^T L = \mathbf{0}_N^T$ gives:

$$\sum_{i=1}^N H_i \ddot{x}_i = \sum_{i=1}^N H_i \xi_i$$

Defining $E_1(t) = \sum_{i=1}^N \nabla f_i(x_i, t)$, its time derivative is:

$$E_2(t) = \dot{E}_1(t) = \sum_{i=1}^N \left(H_i \dot{x}_i + \frac{\partial \nabla f_i}{\partial t} \right)$$

Differentiating $E_2(t)$ and substituting (8) yields:

$$\begin{aligned} \dot{E}_2(t) &= \sum_{i=1}^N \left(H_i \ddot{x}_i + \dot{H}_i \dot{x}_i + \frac{\partial^2 \nabla f_i}{\partial t^2} \right) \\ &= \sum_{i=1}^N \left(H_i \xi_i + \dot{H}_i \dot{x}_i + \frac{\partial^2 \nabla f_i}{\partial t^2} \right) \\ &= \sum_{i=1}^N \left[H_i \left(-H_i^{-1} \left(c_3 \nabla f_i + c_4 \left(H_i \dot{x}_i + \frac{\partial \nabla f_i}{\partial t} \right) \right. \right. \right. \\ &\quad \left. \left. \left. + \dot{H}_i \dot{x}_i + \frac{\partial^2 \nabla f_i}{\partial t^2} \right) \right) + \dot{H}_i \dot{x}_i + \frac{\partial^2 \nabla f_i}{\partial t^2} \right] \\ &= -c_3 \sum_{i=1}^N \nabla f_i - c_4 \sum_{i=1}^N \left(H_i \dot{x}_i + \frac{\partial \nabla f_i}{\partial t} \right) \\ &= -c_3 E_1(t) - c_4 E_2(t) \end{aligned}$$

Consider the Lyapunov function $W(t) = \frac{1}{2} c_3 \|E_1(t)\|^2 + \frac{1}{2} \|E_2(t)\|^2$. Its time derivative is:

$$\begin{aligned} \dot{W}(t) &= c_3 E_1^T \dot{E}_1 + E_2^T \dot{E}_2 \\ &= c_3 E_1^T E_2 + E_2^T (-c_3 E_1 - c_4 E_2) \\ &= -c_4 \|E_2(t)\|^2 \leq 0 \end{aligned}$$

By Barbalat's Lemma, $\dot{W}(t) \leq 0$ implies $\lim_{t \rightarrow \infty} E_2(t) = \mathbf{0}$, which drives $\lim_{t \rightarrow \infty} E_1(t) = \mathbf{0}$. By Lemma 2, the swarm asymptotically tracks $x^*(t)$.

Step 3: Boundedness and Singularity Avoidance ($t \rightarrow (t_0 + T)^-$).

From the differential inequality (13), the decay of the Lyapunov function is bounded by:

$$V_1(t) \leq \frac{1}{\rho(t)^2} V_1(t_0) = \left(\frac{t_0 + T - t}{T} \right)^{2h} V_1(t_0)$$

Since $\frac{1}{2} \|\tilde{v}_i\|^2 \leq V_1(t)$ and $\frac{1}{4} a_{ij} \|x_i - x_j\|^2 \leq V_1(t)$, the error states satisfy the following bounds:

$$\begin{aligned} \|\tilde{v}_i(t)\| &\leq \sqrt{2V_1(t_0)} \left(\frac{t_0 + T - t}{T} \right)^h \\ \|x_i(t) - x_j(t)\| &\leq 2\sqrt{\frac{V_1(t_0)}{a_{ij}}} \left(\frac{t_0 + T - t}{T} \right)^h \end{aligned}$$

Based on the definition of the time-scaling function, the gains are given by $\mu(t) = c_1 + \frac{h}{t_0 + T - t}$ and $\dot{\mu}(t) = \frac{h}{(t_0 + T - t)^2}$. The limits of the critical high-gain coupling terms in the control law (6a) are computed as:

$$\lim_{t \rightarrow (t_0+T)^-} \|\mu(t) \tilde{v}_i\| \leq \lim_{t \rightarrow (t_0+T)^-} O((t_0 + T - t)^{h-1}) = 0$$

$$\lim_{t \rightarrow (t_0+T)^-} \|\dot{\mu}(t)(x_i - x_j)\| \leq \lim_{t \rightarrow (t_0+T)^-} O((t_0 + T - t)^{h-2}) = 0$$

where the selection of the parameter $h > 2$ rigorously ensures that $h - 1 > 0$ and $h - 2 > 0$, mathematically guaranteeing that the exponential decay of the tracking errors strictly dominates the algebraic growth of the time-varying gains.

Furthermore, the condition $V_1(t) \leq V_1(t_0)$ ensures that the APF gradient Φ_i remains globally bounded. Concurrently, the adaptive update laws (5) and (7) guarantee that $\alpha_{ij}(t)$ and $\beta_i(t)$ are monotonically non-decreasing and strictly bounded by α^* and β^* . Consequently, since all constitutive components of the actual control input $u_i(t)$ in (6a) either asymptotically converge to zero or remain strictly bounded, the control singularity is successfully circumvented:

$$\limsup_{t \rightarrow (t_0+T)^-} \|u_i(t)\| < \infty$$

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